Exponential growth and decay

ANSWER KEY

Core Concepts: Exponential growth and decay

1 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(2)$.

a. 15 ✓

b. 21

c. 12

d. 19

Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

2 Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

3 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(3)$.

a. 21 ✓

b. 27

c. 18

d. 25

Substitute $x = 3$: $f(3) = 6(3) + 3 = 21$.

5 Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

6 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Graphical & Geometric Interpretation

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7 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(4)$.

a. 27 ✓

b. 33

c. 24

d. 31

Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

8 Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

9 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(5)$.

a. 33 ✓

b. 39

c. 30

d. 37

Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

11 Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

12 A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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